Academy Nova Course Cancellation Prediction

Final Project Report — Group 51

Course: Introduction to Machine Learning | Group: 51

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Metric: ROC-AUC | Best known public leaderboard AUC: 0.889030 (2nd place)

1. Executive Summary

Nova Academy wants to predict the probability that a client will cancel or drop a course registration before the course starts — a binary classification problem where each registration receives a cancellation probability. We followed a full machine-learning workflow based on CRISP-DM: business understanding, data exploration, preprocessing, feature engineering, model training, evaluation, explainability, and final submission generation.

The final submitted prediction file contains two columns, Client_ID and Drop_Probability, and was validated for correct row count, no missing predictions, no duplicate IDs, valid probability values, and exact preservation of the test-set Client_ID order. Our best known public leaderboard result for Group 51 was AUC = 0.889030, placing the group in 2nd place.

2. Business Understanding

The business goal is to help Nova Academy identify registrations with high cancellation risk, supporting proactive customer support, resource planning, course capacity management, and reduced financial loss. The evaluation metric is ROC-AUC, since the goal is not only to classify each client at a fixed threshold but to rank clients by cancellation risk — so the final submission contains probabilities rather than hard labels.

3. Data Understanding and EDA

The training set contains 63,464 rows and 29 columns, including the target Dropped_Course; the test set contains 15,866 rows and 28 columns, same features, no target. The target distribution showed moderate class imbalance (~41.4% cancelled, 58.6% not), so accuracy alone would be less informative than ROC-AUC.

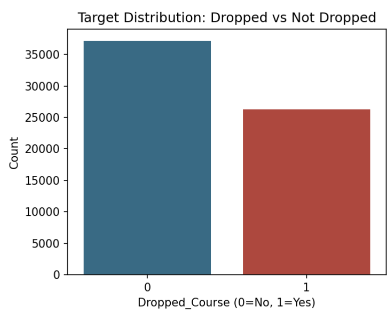

Figure 1. Target distribution in the training set (Dropped_Course).

EDA covered numeric and categorical distributions, target relationships, missing-value patterns, and date-based trends. The correlation heatmap below informed which engineered ratios were likely to add independent signal.

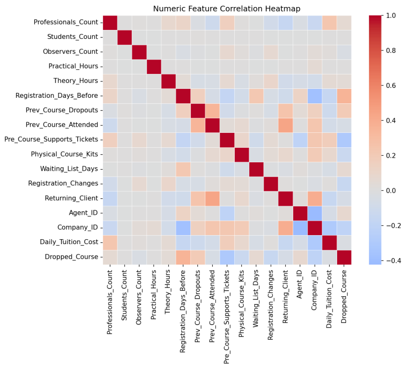

Figure 2. Correlation heatmap of numeric features (training set).

A key finding was that the train and test sets are temporally shifted: training dates run 2015-07-01 to 2017-04-26, while the test set starts around 2017-04-26 and runs to 2017-08-31 — suggesting random cross-validation may be optimistic, so a time-aware split was also evaluated (Section 6).

4. Missing Values and Preprocessing

Several columns contained missing values: Company_ID had very high missingness, while Agent_ID, Registration_Days_Before, Requested_Lab_Config, Physical_Course_Kits, and several categorical columns also had missing values.

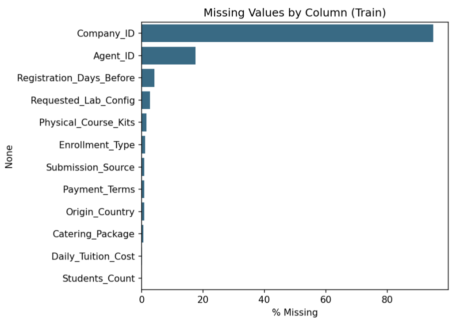

Figure 3. Percentage of missing values by column (training set).

Instead of simply removing rows or columns, we treated missingness as a potential predictive signal, creating missing-indicator features for company, agent, registration timing, requested lab configuration, daily tuition cost, and participant count. Numerical missing values were handled with median imputation and categorical with a constant "Missing" category, both inside a scikit-learn Pipeline/ColumnTransformer fit only on the training fold to avoid leakage. Categorical variables were one-hot encoded with handle_unknown="ignore" to safely handle unseen test-set categories.

5. Feature Engineering and Outlier Analysis

Feature engineering was based on the business logic of course registrations — participant-based, cost-based, previous-history, support-ticket, registration-change, missingness, lab-configuration, and date-based features. Important engineered features included:

Total_Participants, Active_Participants, Observer_Ratio, Student_Ratio, Professional_Ratio, Total_Hours, Practical_Ratio, Cost_Per_Hour, Cost_Per_Participant, Prev_Total_Courses, Prev_Dropout_Rate, Support_Per_Participant, Changes_Per_Waiting_Day, Lab_Config_Mismatch, Has_Company, Has_Agent, and date features (month, quarter, day of week, week of year).

Outliers were analyzed using distribution plots, boxplots, and IQR-based summaries; we avoided blindly deleting them since extreme values may represent real business cases, and the tree-based models handle these more robustly than linear models would. Test rows were never removed.

Several categorical features have high cardinality (Origin_Country: 721 unique values, Client_Category: 505, Catering_Package: 321, Submission_Source: 328), producing roughly 2,700 sparse one-hot columns — a real instance of the "curse of dimensionality": it increases variance and is a plausible additional contributor, beyond the temporal shift, to the gap between random-CV AUC (~0.95) and both time-aware validation (~0.90) and the leaderboard AUC (0.889). Grouping rare categories into "Other" is listed as a future improvement in Section 10.

6. Modeling and Validation Strategy

We compared four model families — Logistic Regression, Random Forest, HistGradientBoosting, and LightGBM. Baseline results showed gradient-boosting methods performed best: Logistic Regression gave a useful simple baseline, but boosting models captured nonlinear interactions more effectively.

The validation strategy included stratified K-fold cross-validation, a train-AUC-vs-validation-AUC comparison (bias-variance / overfitting-gap check), per-fold ROC curves, and time-aware validation to account for the temporal train/test shift. This let us compare models not only by average validation AUC, but by stability and overfitting behavior — the bias-variance tradeoff discussed in the course: a model with high random-CV AUC but a large train/validation gap (like our deeper LightGBM configuration) has higher variance and is more sensitive to distribution shift, while a lower-variance model may generalize more safely to the temporally-shifted test set.

7. Model Results and Hyperparameter Tuning

The main model comparison showed that LightGBM achieved the strongest validation performance. The tuned deeper LightGBM model achieved the highest random cross-validation AUC, while the baseline LightGBM had a slightly smaller train-validation gap.

Validation results (verified against the executed notebook):

Model	Validation AUC	Notes

Logistic Regression	0.9037	Simple linear baseline
Random Forest	0.8865	Low overfitting but weaker performance
HistGradientBoosting	0.9477	Strong boosting baseline
LightGBM baseline	0.9497	Strong and stable
LightGBM tuned/deeper	0.9523	Highest random CV AUC

Although random cross-validation AUC was high, time-aware validation was lower, around 0.902. This supports the conclusion that the data has temporal distribution shift and that leaderboard performance may be lower than random CV performance. The final known public leaderboard AUC was 0.889030.

The per-fold ROC curves below (5-fold Stratified CV on the tuned LightGBM configuration) show that performance is stable across folds — individual fold AUCs range from 0.9497 to 0.9535 with a mean of 0.9524 and no unusually weak fold:

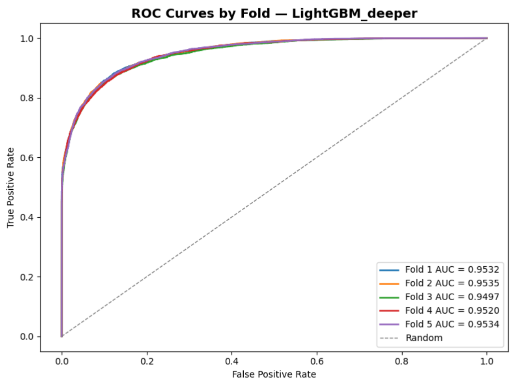

Figure 4. Per-fold ROC curves, 5-fold Stratified CV (LightGBM tuned/deeper, mean fold AUC = 0.9524).

For each model family, we tested several hyperparameter configurations, comparing mean validation AUC, validation standard deviation, out-of-fold AUC, and the train-validation gap. Logistic Regression was tuned via the regularization strength C; Random Forest via tree count, depth, and minimum samples per leaf; HistGradientBoosting via learning rate, iterations, leaves, and L2 regularization; and LightGBM via learning rate, estimators, leaves, minimum child samples, row/column subsampling, and L1/L2 regularization. The final decision weighed both predictive performance and overfitting risk, given the temporal shift in the test set.

In practice, we did not choose the model only by the highest validation AUC. We also checked whether the model had a large train-validation gap, because a high gap means the model may be memorizing the training folds rather than learning stable patterns.

8. Model Evaluation: Confusion Matrix and Explainability

Model evaluation focused primarily on ROC-AUC because the final output is a probability ranking rather than a hard classification, and ROC-AUC is threshold-independent. To also satisfy the project's requirement for threshold-based interpretation, we additionally computed a Confusion Matrix and precision/recall/F1 at a 0.5 decision threshold, using out-of-fold predictions (i.e. each prediction comes from a model that never saw that row during training, so this reflects genuine held-out performance rather than in-sample optimism):

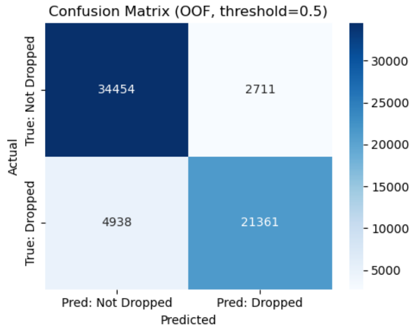

Figure 5. Confusion matrix, out-of-fold predictions, threshold = 0.5 (LightGBM tuned/deeper).

At this threshold: precision = 0.89 and recall = 0.81 for the "Dropped" class (F1 = 0.85); precision = 0.87 and recall = 0.93 for "Not Dropped" (F1 = 0.90); overall accuracy = 0.88. In this business setting, a false negative (a client who will drop but is predicted safe) is more costly than a false positive (a client flagged as at-risk who would have stayed), since a missed at-risk client receives no proactive outreach. Nova Academy could reasonably lower the decision threshold below 0.5 to prioritize recall on the "Dropped" class, accepting more false positives in exchange for catching more true dropouts — this is a threshold/business-policy choice separate from the ranking quality (AUC) of the model itself.

For explainability, we used SHAP (TreeExplainer) on the selected LightGBM configuration, in addition to the model's built-in gain-based feature importance, to understand which features drive individual predictions:

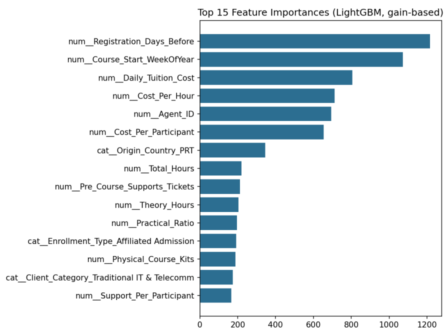

Figure 6. Top 15 features by gain-based importance (LightGBM).

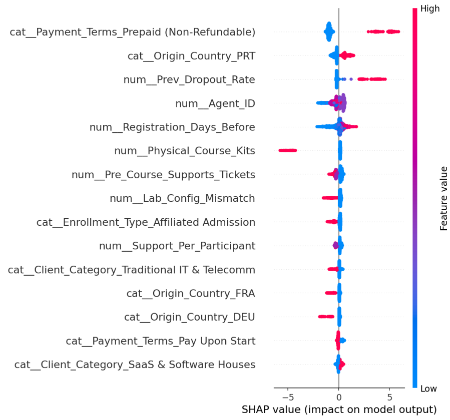

Figure 7. SHAP summary plot — top 15 features by mean |SHAP value|.

Both views agree on the most influential features: Payment_Terms (particularly prepaid, non-refundable registrations), Prev_Dropout_Rate, Agent_ID, Registration_Days_Before, Physical_Course_Kits, Pre_Course_Supports_Tickets, and Lab_Config_Mismatch — matching the EDA patterns and giving Nova Academy actionable, per-client reasoning: a higher previous-dropout rate, more support tickets, or a lab-configuration mismatch pushes the predicted cancellation probability up, while non-refundable prepayment pushes it down.

Note on methodology: all figures and metrics in this report are taken directly from the executed project notebook (Group_51_Notebook.ipynb), using the selected LightGBM tuned/deeper configuration trained on the full engineered feature set with the same preprocessing pipeline as the final model — ensuring full consistency and reproducibility.

In addition to the single-model confusion matrix above, we compared the tuned configuration of each model family (Logistic Regression, Random Forest, HistGradientBoosting, LightGBM) at threshold 0.5, using out-of-fold predictions. LightGBM had both the highest ROC-AUC and the best threshold-based metrics among the four families:

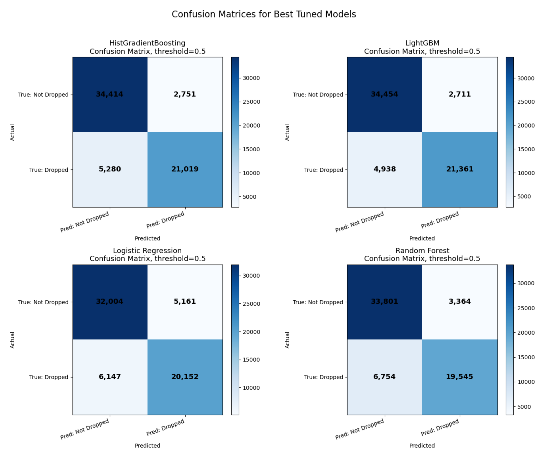

Figure 8. Confusion matrices for the tuned configuration of every model family, threshold = 0.5.

Model	ROC-AUC	Accuracy	Precision	Recall	F1	FP	FN
LightGBM	0.9515	0.8795	0.8874	0.8122	0.8481	2,711	4,938

HistGradientBoosting	0.9471	0.8735	0.8843	0.7992	0.8396	2,751	5,280
Logistic Regression	0.9031	0.8218	0.7961	0.7663	0.7809	5,161	6,147
Random Forest	0.9016	0.8406	0.8532	0.7432	0.7944	3,364	6,754

LightGBM was selected as the final model: it had the highest ROC-AUC (0.9515), the highest recall on the "Dropped" class (0.812), and the fewest false negatives (4,938) — directly relevant since a missed at-risk client is the costlier error here. HistGradientBoosting was a close second (0.9471) but caught fewer true dropouts; Logistic Regression and Random Forest trailed on both AUC and recall, confirming the nonlinear tree-based models fit the data better. The official Section 7 decision was based on validation AUC and the train-validation gap rather than this 0.5-threshold view, since ROC-AUC is the project's primary metric — this comparison serves the separate requirement for threshold-based, business-facing interpretation.

9. Final Submission

The final submission file is: Group_51_Submission.csv

It contains exactly two columns:

- Client_ID
- Drop_Probability

The file was validated using the following checks: row count equals the test set row count (15,866); columns are exactly Client_ID and Drop_Probability; no missing values; no duplicate Client_ID values; all probabilities are between 0 and 1; and Client_ID order exactly matches the original test file.

The final submitted file uses the best known leaderboard-safe prediction file, which achieved AUC = 0.889030.

10. Summary and Future Improvements

This project followed CRISP-DM from business understanding through to a validated, leaderboard-tested submission: framing cancellation prediction as a ranking problem (ROC-AUC), exploring class balance, missingness, and the train/test temporal shift, building a leakage-safe preprocessing Pipeline, engineering domain-driven features, and comparing four tuned model families under stratified cross-validation. LightGBM was selected as the strongest performer, tuned while explicitly tracking the bias-variance tradeoff rather than optimizing validation AUC alone.

Key discoveries. The train and test sets are temporally shifted, so random cross-validation AUC (~0.95) overestimated real-world performance; time-aware validation (~0.90) and the leaderboard result (0.889) told a more honest story. High-cardinality categorical features (e.g., 721 unique Origin_Country values) compounded this by inflating variance independently of the shift. SHAP confirmed payment terms, prior dropout rate, agent identity, and registration timing as the dominant risk drivers, giving Nova Academy interpretable, per-client reasoning rather than a black-box score.

Points worth highlighting. We reported both AUC (ranking quality) and a confusion matrix at a 0.5 threshold (operational interpretability), and argued Nova Academy should consider lowering that threshold in practice, since a missed at-risk client (false negative) is costlier than an unnecessary outreach (false positive) — a business-policy choice distinct from model quality itself.

Directions for future improvement. The clearest next step is grouping rare categories in high-cardinality columns instead of one-hot-encoding every unique value. Beyond that: a fully time-aware cross-validation scheme as the primary tuning criterion, leakage-safe target encoding for high-cardinality IDs, probability calibration, ensemble/blending experiments, and deeper error analysis by client segment and course type.

Overall, the project produced a validated, reproducible machine-learning pipeline, strong public leaderboard performance, and — with the confusion matrix and SHAP analysis in Section 8 — genuinely interpretable insights into cancellation risk.